

Investigation: Slim Jim Antenna (2m Band)

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Target Frequency: 145 MHz

Antenna Type: Slim Jim (Folded End-Fed Half-Wave)

1. Introduction

This investigation documents the physical design, construction, tuning, and deployment of a Slim Jim antenna for the 2-meter band. The goal was not only to achieve low SWR, but to develop an intuitive and measurable understanding of how geometry influences impedance, resonance, and matching behavior. The antenna was iteratively tuned using a NanoVNA, with measurements taken both at the feedpoint and through a 15 m coaxial feedline to evaluate real-world system performance.



2. Objectives

- Validate Slim Jim geometry for ~145 MHz operation
- Understand impedance sensitivity to:
 - Feedpoint position
 - Stub length
 - Capacitive gap
- Evaluate deployed performance through longer coax
- Develop a mechanically stable and repeatable construction method

3. Design Overview

3.1 Electrical Design Parameters (Realized Build)

Parameter	Value
Antenna Height	58"
Total Wire Length	~116"
Wire Type	14 AWG THHN
Parallel Spacing	0.5"
Capacitive Gap	1.0" (initial), 0.5" (final)
Stub Length	18" (initial), 18.5" (final)
Feedpoint	1.5" (initial), 3" (final)

*Note:
The antenna height (~58") represents the folded structure.
The total RF path (~116") is distributed along both conductors.

4. Design Rationale

4.1 Free-Space Reference (Theoretical Measurements)

- At 145 MHz:
- $\lambda \approx 81.5"$ (full wavelength)
 - $1/2 \lambda \approx 40.75"$ (radiating section)
 - $1/4 \lambda \approx 20.4"$ (matching stub)
 - $3/4 \lambda \approx 61.1"$ (total electrical length)

*Note:
These values serve as a theoretical reference, not final dimensions.

4.2 Realized Electrical Lengths (Measured Build)

Section	Length
Radiating Section	~39"
Matching Stub	18.5"
Effective Electrical Length	~57.5"

This corresponds to:
Effective wavelength scaling factor $\approx 57.5 / 61.1 \approx 0.94$

4.3 Physical vs Electrical Length (Critical Understanding)

The Slim Jim antenna must be understood between its **electrical length** and its **physical geometry**. These are not equivalent. Electrically, the antenna behaves as a 3/4 wavelength structure, with an effective length of about **57.5" not 61.1"** in the realized build. This length represents the actual path that RF current follows along the conductor. In contrast, the physical height of the antenna is approximately 58 inches including the cap gap, which reflects only the folded geometry of the structure rather than the full electrical path.

Because the antenna is folded, the total conductor length is roughly **116 inches**, or about twice the physical height. The RF current does not travel straight up a single vertical element; instead, it follows a distributed path: it moves upward along one conductor, transitions across the top of the antenna, and then travels downward along the parallel conductor. This folded current path allows the antenna to achieve the electrical behavior of a 3/4 wavelength structure within a more compact physical form.

As a result, the **$1/2\lambda$ radiating section** and **$1/4\lambda$ matching stub** should not be interpreted as separate vertical segments stacked in space. Rather, they are **electrical regions** along the continuous conductor path, shaped and influenced by the antenna's folded geometry and coupling between elements.

This distinction is critical for both design and analysis. Misinterpreting electrical length as physical height leads to incorrect assumptions about resonance, impedance, and overall antenna behavior.

4.4 Parallel Wire Spacing (0.5")

The parallel section of the Slim Jim functions as a **two-wire transmission line**, where the characteristic impedance (Z_0) is not determined by conductor size alone, but by the **ratio of spacing to conductor diameter**.

- $Z_0 \propto \text{spacing} / \text{diameter}$
- Larger diameter - lowers Z_0 (more capacitance)
- Larger spacing - raises Z_0

This explains why constructions using **copper tubing**, larger diameter, also use **much wider spacing**. The wider gap compensates for the increased conductor size, keeping the spacing-to-diameter ratio—and therefore Z_0 —within a usable range. (Section E5D)

In this design:

- 14 AWG wire (~1.6 mm diameter)
- ~0.5" spacing produces a similar ratio to common copper pipe builds
- Resulting in a **moderate Z_0** , suitable for impedance transformation

This was validated during tuning:

- Lower feedpoint - ~30 Ω
- Raising feedpoint increased resistance
- ~3" feedpoint - ~56 Ω with low reactance

This produces an impedance profile that passes through the 50 Ω region, which is the goal of the matching stub.



4.5 Stub Length (~18.5")

The lower section of the Slim Jim acts as a **quarter-wave matching stub**, transforming the high impedance at the open end into a usable feedpoint impedance along its length. In free space, a quarter wavelength at 145 MHz would be approximately **20.4 inches**, but the realized build uses a shorter length of **18.5 inches**.

This reduction is expected and results from several physical effects that electrically “slow down” the wave:

- Insulation (THHN) - lowers velocity factor
- Mutual coupling between the parallel conductors
- Capacitive loading at the cap gap

Together, these effects shorten the required physical length for the stub to behave as a quarter-wave section.

From a tuning perspective, the stub length primarily controls **resonance and reactance**:

- Increasing stub length - lowers resonant frequency
- Increasing stub length - reduces inductive reactance (+j)
- Shortening the stub - pushes the system more inductive

In practice, an initial 18" stub was accounted for and slightly short (inductive), and increasing it to 18.5" brought the system closer to resonance, as confirmed by the shift from +j reactance toward near-zero.

4.6 Capacitive Gap

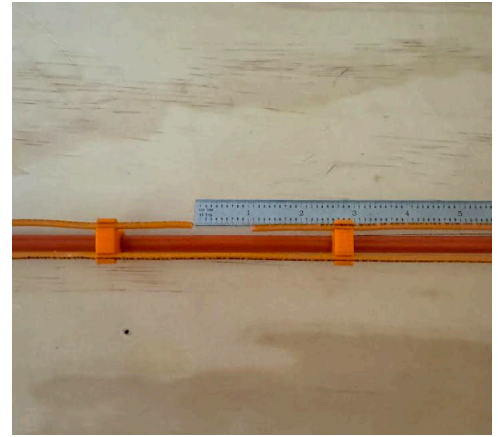
The cap gap at the bottom of the Slim Jim introduces a capacitive element into the system, which plays an important role in fine-tuning the antenna's reactance. This gap exists at the open end of the matching stub, where electric fields concentrate, making it an effective point for capacitive loading.

The size of the gap directly controls how much capacitance is introduced:

- Larger gap - less capacitance - system remains more inductive
- Smaller gap - more capacitance - cancels inductive reactance

Unlike the stub length, which primarily sets the overall resonance, the capacitive gap acts as a **fine adjustment mechanism**. It allows small corrections to bring the antenna closer to a purely resistive impedance after the major tuning has been completed.

Reducing the gap from **1.0" to 0.5"** added capacitance, which helped offset the remaining inductive reactance and move the system closer to resonance, as reflected in the measured impedance.



4.7 Feedpoint Position

The feedpoint location along the stub is the **primary control of input resistance**, as it determines where along the impedance transformation the antenna is sampled. The matching stub creates a continuous impedance gradient, ranging from low impedance near the shorted end to high impedance near the open end.

As the feedpoint is moved:

- Lower position - samples lower impedance (~20–30Ω)
- Higher position - samples higher impedance (~50–70Ω)

This behavior follows directly from transmission line theory, where impedance varies along the length of a quarter-wave section. Rather than changing the antenna itself, adjusting the feedpoint simply selects a different point along this transformation.

During tuning, raising the feedpoint from **1.5" to 3"** increased the measured resistance toward the desired 50Ω region, making this the most effective and immediate adjustment for achieving a proper match.



5. Mechanical Design & Construction

5.1 Structural Support

A fiberglass driveway marker at 75" was used as a boom for the wire and for mounting structures.

- Electrically neutral
- Lightweight and rigid

5.2 3D Printed Spacers

- Maintain 0.5" spacing
- Snap-fit onto rod
- Dual-lobe design

Ensures consistent impedance behavior

5.3 Air Choke (Common Mode Suppression)

An air-wound choke was added at the feedpoint to suppress **common-mode currents** on the feedline. Without this, the coax can become part of the antenna system, distorting the radiation pattern and affecting impedance measurements.

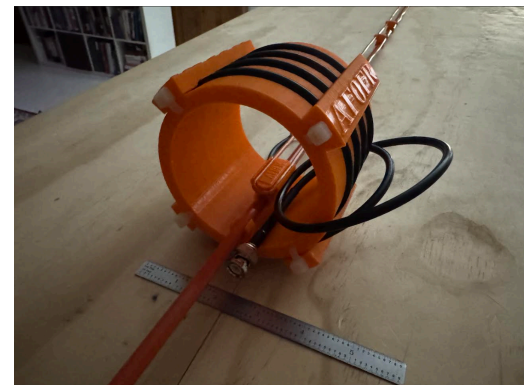
- Diameter: 4 in
- Spacing: 0.25 in

The design is based on two key principles:

- Larger diameter → increases inductance per turn, improving choking impedance
- Turn spacing → reduces self-capacitance, preventing the choke from becoming resonant

Result:

- High impedance to common-mode currents
- Prevents feedline radiation
- Stabilizes measurements



6. Measurement Results

6.1 Initial Build

- Impedance: $33.28 + j28.6 \Omega$
- SWR: 2.6

Interpretation

- Low resistance $\rightarrow$ feedpoint too low
- $+j$ reactance $\rightarrow$ antenna electrically short

6.2 Tuned Configuration

Adjustments:

- Feedpoint: 1.5" $\rightarrow$ 3"
- Stub: 18" $\rightarrow$ 18.5"
- Gap: 1" $\rightarrow$ 0.5"

Results:

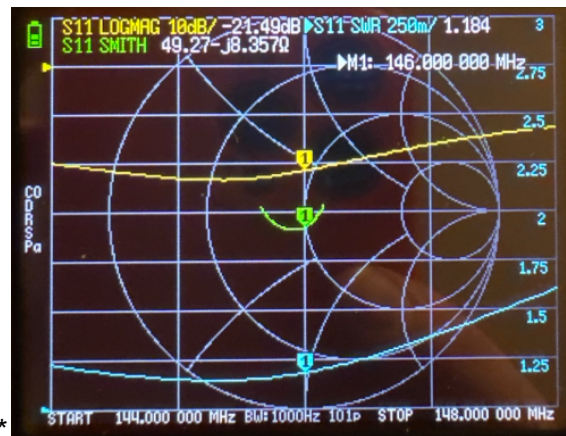
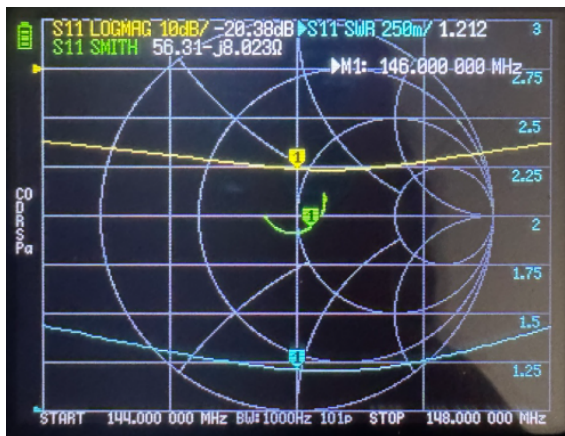
- Impedance: $56.31 - j8.02 \Omega$
- SWR: 1.21

Interpretation

- Resistance moved toward 50Ω
- Reactance shifted inductive $\rightarrow$ slightly capacitive

Confirms:

- Stub controls resonance
- Gap cancels residual reactance



*Note:

These measurements show the final tuned antenna before deployment (shown in the first image), taken after permanent soldering. During initial tuning, alligator clips were used to adjust the feedpoint position, which allowed for a slightly improved match (shown in the second image).

6.3 Deployed, Transmitter Measurement (15 m Coax)

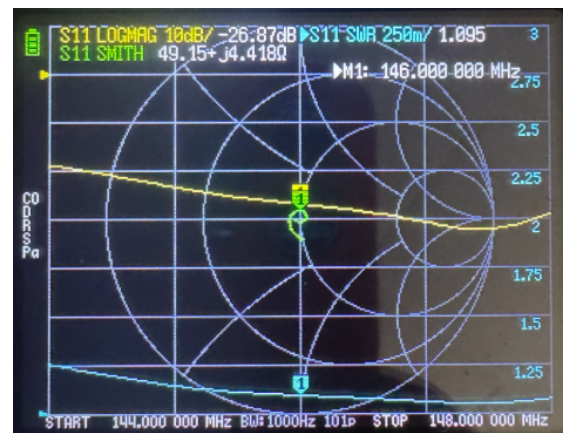
- Impedance: $49.15 + j4.42 \Omega$
- SWR: 1.095

*Note:

When measured through the 15 m coaxial feedline, the antenna appears closer to an ideal 50Ω match compared to the direct feedpoint measurement. This improvement is not due to a change in the antenna itself, but rather the effect of the transmission line.

The coax introduces two key effects:

- **Impedance rotation:**
The reactive component is transformed along the length of the line, shifting the measured impedance (e.g., from slightly capacitive to slightly inductive).
- **Loss masking:**
Attenuation in the coax reduces the magnitude of reflected power, which artificially lowers the measured SWR and makes the match appear better than it truly is at the antenna.



8. System Behavior Summary

Stage	Resistance	Reactance	SWR	Insight
Initial	33Ω	+j28	2.6	Too short
Tuned	56Ω	-j8	1.21	Acceptable resonance
Deployed	49Ω	+j4	1.09	Coax transformed
Reference	50Ω	+0	1	Ideal

9. Key Insights

- 1. Feedpoint dominates resistance
- 2. Stub length controls resonance
- 3. Gap fine-tunes reactance
- 4. Geometry consistency is critical
- 5. Choke stabilizes system behavior

10. Conclusions

- The final antenna achieved near-ideal impedance matching after iterative tuning, with measurements approaching 50Ω and minimal reactance. This confirms that the chosen geometry and tuning approach successfully produced a well-matched system for the target frequency.
- Throughout the process, it became clear that real-world antenna dimensions do not directly align with ideal free-space calculations. Factors such as insulation, conductor diameter, coupling between elements, and capacitive loading all contribute to shortening the required physical lengths. As a result, practical designs must rely on both theoretical guidance and empirical adjustment.
- A key takeaway is that electrical length must be understood along the conductor path, not the physical height of the antenna. The folded geometry of the Slim Jim allows it to achieve the behavior of a longer electrical structure within a compact form, which can be misleading if interpreted purely from physical dimensions.
- Finally, measurements taken through a long feedline demonstrated that transmission line effects can mask the true antenna performance. Impedance rotation and loss can make the system appear better matched than it is at the feedpoint, reinforcing the importance of validating both direct and system-level measurements.

11. Appendix

- 70cm measurement in the deployed environment.
 - Impedance: $46.28 + j5.887 \Omega$
 - SWR: 1.156



- QSO with KF0WGM on 147.195 MHz: Potosi 43.67mi
QSA was about a 4, I just needed to talk on the mic a bit closer.

